

Self Supervised RL Curriculum Report

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Self Supervised Curriculum Generation for Autonomous Reinforcement Learning

Executive assessment

The most useful organizing distinction is between two kinds of self-supervised curriculum generation.

1. **Evaluation-driven task selection.** The learner measures its current competence on candidate latent states and chooses additional training data where those measurements suggest high expected improvement.
2. **Reachability-driven autonomous exploration.** The learner discovers and masters states that are reachable from a starting state, often by expanding a frontier of controllable states.

The first category is closest to the model of interest. It contains the canonical empirical algorithms, but almost no formal guarantees for the full deep-RL setting. The second category contains the strongest statistical theory, but its objective is usually to discover reachable states or learn goal-conditioned policies rather than maximize aggregate performance over an exogenous latent-state distribution.

The central conclusion is therefore a theory gap, not an absence of theory: existing results provide useful components for the proposed model, but no paper in the search gives a general high-probability guarantee for adaptive latent-state sampling from noisy i.i.d. evaluation signals under deep reinforcement-learning dynamics.

The evaluation-conditioned model

Let $z \in \mathcal{Z}$ denote an observable latent state or task descriptor. For each z , the learner can query an i.i.d. evaluation environment and observe a performance signal for its current policy π_t . It then chooses a latent state z_t from which to collect additional training data. The curriculum policy is adaptive because z_t depends on the history of evaluation outcomes and training data.

A generic objective is aggregate performance under a target distribution ν :

$$J(\pi) = \mathbb{E}_{z \sim \nu}[V^\pi(z)].$$

This formulation separates four objects that are often conflated in the curriculum literature:

Component	Role in the model	Closest literature	Main unresolved issue
Evaluation	Estimate current competence at z	TSCL, GoalGAN, ALP-GMM, CURIOUS	Evaluation noise and adaptive reuse of measurements
Curriculum query	Select the next latent state for training	Learning-progress and frontier methods	Which score is statistically tied to future improvement
Training update	Change the policy using data from the selected state	All methods	Nonstationary learning and transfer across states

Component	Role in the model	Closest literature	Main unresolved issue
Final objective	Improve average performance over ν	Contextual curriculum methods	Most theory instead targets reachability or one target task

The model is broader than goal-conditioned exploration. A latent state may specify a goal, initial-state distribution, environment parameter, reward condition, or other observable context. The key requirement is that the learner can evaluate performance conditioned on that state before deciding where to train next.

Canonical evaluation-driven algorithms

The 2020 survey by Narvekar et al. provides the standard vocabulary: curriculum generation creates intermediate tasks, sequencing chooses among them, and transfer determines how knowledge moves between tasks [Nar20]. Under the present lens, the most important methods are those whose sequencing rule is driven by the learner’s own evaluation signal.

Method	Training query	Evaluation signal	Formal guarantee	Empirical finding
TSCL	Discrete subtasks	Slope or absolute slope of task performance	None	Minecraft curriculum was reported to learn about an order of magnitude faster than uniform sampling [Mat17]
GoalGAN	Continuous goals	Success rate in an intermediate band	None	Expands coverage of reachable goals and avoids large unlearnable regions [Hel17]
Reverse curriculum	Start states near a goal	Return or success probability from each start state	No new PAC result	Solved sparse-reward navigation and manipulation tasks that failed under standard sampling [Flo17]
ALP-GMM	Continuous environment parameters	Absolute local reward change	None	Robust to unlearnable regions and high-dimensional task parameters [Por19]
CURIUS	Modular goals	Absolute learning progress	None	Improved learning and recovery after competence loss in modular robotic tasks [Col18b]

Method	Training query	Evaluation signal	Formal guarantee	Empirical finding
Value disagreement	Goals in a multi-goal task	Ensemble disagreement in goal-conditioned values	None	Improved sample efficiency over HER, ALP-GMM, RIAC, and GoalGAN on sparse-reward tasks [Zha20b]

These methods share a common heuristic: do not spend most training effort on states that are already mastered or effectively impossible. Their signals differ in what they treat as evidence of learnability.

- **Learning progress** estimates whether competence is changing, as in TSCL, ALP-GMM, and CURIOS.
- **Intermediate success** selects tasks whose success probability is neither near zero nor near one, as in GoalGAN and reverse curriculum generation.
- **Epistemic uncertainty** uses disagreement or policy uncertainty as a proxy for the frontier of current competence, as in value disagreement and newer uncertainty-based methods.
- **Target relevance** adds information about whether a candidate task helps a specified final objective, as in DISCOVER and ProCuRL-Target.

None of these papers proves that its score is an unbiased or even consistent estimate of marginal improvement in the general deep-RL setting. The observed gains are therefore algorithm-specific empirical findings rather than consequences of a general curriculum theorem.

The strongest formal theory

Autonomous exploration and goal-conditioned reachability

The closest formal literature is autonomous exploration in reward-free or goal-conditioned Markov decision processes. Here the learner selects states or goals to discover and learns policies for reaching them.

AdaGoal selects goals that are estimated to be reachable but uncertain. In tabular MDPs, it gives a $\tilde{O}(L^3 S A \epsilon^{-2})$ exploration bound and is nearly minimax under a reset-action assumption. A linear-mixture version gives a corresponding function-approximation result [Tar21]. The algorithm’s frontier logic is highly relevant: candidate states are chosen because they are neither too easy nor too difficult according to an uncertainty estimate.

Value-Aware Autonomous Exploration makes policy evaluation an explicit part of the algorithm. It alternates between optimistic model-based policy computation and empirical evaluation of whether a candidate goal can be reached at the predicted cost [Cai22]. The paper also provides a lower bound, showing near-minimax behavior when the number of incrementally controllable states grows moderately with the reachability radius. This is the closest formal precedent for using an evaluation signal to decide whether a state should enter the learned curriculum.

Layered Autonomous Exploration improves the picture by defining reachability layers and expanding only from states already certified as controllable [Che23]. Under a layer-identifiability condition, the resulting sample complexity matches the autonomous-exploration lower bound up to logarithmic factors

and avoids dependence on the total state-space size. The result even covers countably infinite state spaces. Its curriculum interpretation is clear: first consolidate an inner layer, then use it to discover the next layer. DisCo provides an earlier model-based version of the same idea, interleaving state discovery, optimistic evaluation, and policy consolidation [Tar20c]. Its guarantee has a worse dependence on the exploration radius and branching factor than the later layered method, but it supplies useful empirical evidence that evaluation-conditioned frontier expansion can improve discovery rates.

These results are powerful but differ from the evaluation-conditioned latent-state model in three important ways:

- They usually treat latent states as physical states or goals inside one fixed MDP, not as contexts indexing i.i.d. evaluation environments.
- Their objective is to discover reachable states and learn goal-conditioned policies, not directly maximize average performance under a target distribution ν .
- Their guarantees rely on tabular or structured transition models, reset actions, reachability conditions, and carefully defined shortest-path objectives.

Sample collection as a separate layer

A useful decomposition separates curriculum design from physical data collection. GOSPRL accepts an adaptive set of desired state-action sample counts and navigates an unknown MDP to fulfill them. Its cost is roughly the requested number of samples multiplied by an exploration-diameter overhead, up to lower-order terms [Tar20b].

This result suggests a modular theory for the proposed model:

1. An evaluation-driven curriculum rule decides which latent states or state-action regions deserve data.
2. A sample-collection procedure converts those requests into actual online trajectories.
3. A learning theorem bounds the improvement in aggregate performance.

The literature has developed the second component much more fully than the first and third.

Curriculum transfer theory

ROLLIN studies a different but complementary question: when a useful curriculum is already available, can sequentially solving related tasks reduce the complexity of learning a difficult target task? Under Lipschitz reward variation and sufficiently small changes between adjacent contexts, the authors show that warm starts and roll-in distributions can turn a hard single-task problem into a sequence of easier problems [Li22b].

This is a guarantee for a structured curriculum, not for discovering the curriculum from evaluation data. It supports the idea that adaptive state selection can help only when neighboring latent states share useful information, but it does not identify which state should be selected next or quantify the cost of estimating that choice.

Directed exploration toward a target

DISCOVER is the most relevant newer result for target-directed curriculum generation. It selects achieved goals using three signals: achievability, novelty, and relevance to a final target goal. Under linear value functions in a latent feature space, noisy value feedback, geodesic structure, and a positive expansion rate,

it bounds the number of episodes before the target becomes achievable by approximately $\tilde{O}(Dd^2\kappa^{-3})$, independent of the volume of the full goal space [Dia25].

DISCOVER is closer to the proposed model than generic autonomous exploration because it uses the learner’s own estimates to select additional training goals and explicitly analyzes noisy feedback. Its scope remains narrower: the theorem concerns making one target goal achievable, not improving expected performance over an arbitrary latent-state distribution, and it assumes a favorable latent geometry.

Newer evaluation-driven methods

Recent work sharpens the empirical and partial-theory picture.

Learnability rather than regret proxies

Sampling For Learnability argues that common regret approximations in unsupervised environment design often select environments the agent already solves rather than environments where it can improve. It proposes the Bernoulli-variance score

$$p(z)(1 - p(z)),$$

where $p(z)$ is the estimated success rate. This targets the intermittent-success frontier directly. In single-agent navigation and grid-world settings, the method improves robustness over several UED baselines, especially under worst-case evaluation [Rut24].

The conceptual importance is greater than the particular score. It shows that a performance estimate can be more useful than a value-based uncertainty proxy when the curriculum objective is to find tasks that are sometimes solvable. It also exposes a statistical issue: the evaluation signal must distinguish mastery, impossibility, and genuine learnability.

Target-aware proximal curricula

ProCuRL-Target selects tasks that are both in the learner’s zone of proximal development and correlated with tasks from the target distribution [Tza24]. Its formal analysis derives the rule in a simplified REINFORCE and contextual-bandit model and shows logarithmic dependence on target accuracy under that model. The deep-RL algorithm used in experiments is an extension of the analysis rather than a directly covered case.

This paper is useful for the aggregate-objective version of the problem because it explicitly distinguishes learning potential from target relevance. A curriculum can improve on a candidate task without improving the final metric if the candidate is poorly related to the target distribution.

Skill-based Bayesian curriculum models

SEBNs model environment features, latent competencies, and observable task outcomes in a Bayesian network [Hsi25]. The curriculum estimates performance on candidate tasks, including unseen combinations of features, and selects tasks with large predicted changes in competence. This is one of the clearest recent examples of explicitly modeling a latent skill state between evaluation outcomes and task choice.

The method has no global convergence or sample-complexity guarantee. Its contribution is architectural: it makes the mapping from observable evaluation outcomes to latent competencies explicit, which may be useful when the latent state is structured and task descriptors are compositional.

Relative-entropy curriculum selection

READ-C selects start states with high policy uncertainty, measured by relative entropy from an optimal or estimated reference policy [Sat25]. The paper proves convergence of the underlying two-timescale actor-critic process under standard stochastic-approximation assumptions. That theorem does not establish that the curriculum selector is statistically optimal, nor that it minimizes the number of evaluation or training samples needed to improve aggregate performance.

The distinction matters. Convergence of the learner under an adaptive state distribution is weaker than a curriculum guarantee. The latter would need to relate the state-selection rule to improvement in the target objective.

What the theory does and does not establish

Question in the evaluation-conditioned model	Existing support	Remaining gap
Can evaluation identify useful training states?	Frontier and intermediate-success heuristics; VALAE evaluation rounds [Cai22]	No general consistency or regret result for noisy adaptive evaluation
Can the learner discover new useful states?	Autonomous-exploration guarantees [Tar21, Cai22, Che23]	Usually assumes fixed-MDP reachability and reset structure
Can task choice improve aggregate performance?	Contextual curriculum theory and target-aware heuristics [Li22b, Tza24]	No general bound for adaptive deep-RL task selection
Can online interaction fulfill requested data allocations?	GOSPRL provides an online sampling reduction [Tar20b]	Does not determine the optimal allocation itself
Can latent structure reduce search complexity?	Linear-mixture and latent-feature results [Tar21, Dia25]	Strong representation and geometry assumptions
Can curricula converge under adaptive starts?	Two-timescale convergence analysis [Sat25]	No optimality or sample-efficiency guarantee for the selector

The strongest results rely on at least one of four structural assumptions:

- **Reset or generative access**, which makes evaluation and training episodes separable.
- **Reachability structure**, such as a bounded horizon, layered controllability, or a finite branching factor.
- **Smooth task relatedness**, such as Lipschitz reward changes or transferable policy representations.
- **Low-dimensional latent geometry**, such as linear value functions or geodesic paths toward a target.

The user’s model makes the first assumption more plausible through i.i.d. evaluation environments, but it does not by itself solve the other three problems. In particular, i.i.d. evaluation removes temporal dependence from competence estimation; it does not say how competence at one latent state predicts the value of training at another.

A theory target for the proposed model

A formal treatment would need to specify four linked statistical objects.

1. **Evaluation estimation.** For each latent state z , evaluation returns should provide a noisy estimate of $V^{\pi_t}(z)$, with a concentration condition that remains valid under adaptive state queries.
2. **Training response.** The update obtained from training at z_t should be related to changes in performance at other latent states, perhaps through a transfer kernel or a shared representation.
3. **Curriculum objective.** The selector should be judged by improvement in $J(\pi_t)$, not merely by the number of reachable or mastered states.
4. **Nonstationarity and forgetting.** The analysis should allow training at one state to improve, leave unchanged, or damage performance elsewhere.

A useful guarantee would bound the gap between the aggregate performance of an adaptive curriculum and that of the best allocation of training queries, while accounting for evaluation samples, training samples, and online interaction cost. Existing work supplies partial ingredients: empirical frontier scores estimate learnability [Rut24], autonomous exploration supplies reachability guarantees [Tar21, Cai22, Che23], target-aware curricula model transfer relevance [Tza24], and directed exploration gives a latent-geometry bound [Dia25]. No existing result combines all four.

Bottom line

The current literature supports the following view.

- The canonical algorithms are TSCL, GoalGAN, reverse curriculum generation, ALP-GMM, CURIOUS, and value-disagreement sampling [Mat17, Hel17, Flo17, Por19, Col18b, Zha20b]. They are strong empirical baselines for evaluation-driven latent-state selection but lack general statistical guarantees.
- The canonical theory is autonomous exploration and multi-goal SSP [Tar21, Cai22, Che23, Tar20c]. It gives near-minimax guarantees for discovering and mastering reachable states, but does not directly optimize aggregate performance over i.i.d. latent-state environments.
- The most relevant newer work is DISCOVER, which adds target relevance and a latent-feature guarantee [Dia25], together with learnability-based selection, target-aware proximal curricula, Bayesian skill models, and relative-entropy selection [Rut24, Tza24, Hsi25, Sat25]. These methods improve the match to the proposed model but remain partial theories.
- The main open problem is to obtain a high-probability allocation or regret guarantee for adaptive training-state selection when competence is observed through i.i.d. evaluation environments and training has transferable, nonstationary effects across the latent space.

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